

MATH 501: REAL ANALYSIS

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1. AUGUST 25, 2026

Real analysis will be developed through measure theory, Lebesgue integration, and the study of function spaces such as L^p spaces.

1.1. Measures. The goal of a measure is to assign a length, area, or volume to subsets of $\mathbb{R}$, $\mathbb{R}^2$, or, more generally, $\mathbb{R}^n$. Ideally, we would like a function

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

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with the following properties.

- (i) *Countable additivity*: if $E_1, E_2, \dots \subseteq \mathbb{R}^n$ are pairwise disjoint, then

$$m\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} m(E_j).$$

A weaker requirement is finite additivity, in which the identity is required only for finite pairwise disjoint families.

- (ii) *Invariance under rigid motions*: if

$$\tilde{E} = AE + b, \quad A \in O(n), \quad b \in \mathbb{R}^n,$$

then $m(\tilde{E}) = m(E)$. Here A represents a rotation or reflection, and b a translation.

- (iii) *Normalization*:

$$m([0, 1]^n) = 1.$$

Unfortunately, a measure with all these properties cannot be defined on every subset of $\mathbb{R}^n$.

Theorem 1 (Vitali). *Assuming the axiom of choice, there is no countably additive, rigid-motion-invariant map*

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

satisfying $m([0, 1]^n) = 1$.

Proof. It is enough to consider $n = 1$: if such a map existed on $\mathbb{R}^n$, then

$$\mu(E) = m(E \times [0, 1]^{n-1}), \quad E \subseteq \mathbb{R},$$

would have the required properties on $\mathbb{R}$. We therefore write m for this one-dimensional map.

First, $m(\{x\}) = 0$ for every $x \in \mathbb{R}$. Otherwise, translation invariance and countable additivity applied to a countably infinite set in $[0, 1]$ would contradict $m([0, 1]) = 1$. Consequently, $m([0, 1)) = 1$.

Define an equivalence relation on $[0, 1)$ by

$$x \sim y \iff x - y \in \mathbb{Q}.$$

Using the axiom of choice, select one representative from each equivalence class and let V be the set of representatives. Thus, for every $x \in [0, 1)$, there is a unique $y \in V$ such that $x - y \in \mathbb{Q}$.

For $t \in [0, 1)$, define the translate of V modulo 1 by

$$\begin{aligned} V_t &= V + t \pmod{1} \\ &= \{x + t : x \in V, x + t < 1\} \\ &\quad \cup \{x + t - 1 : x \in V, x + t \geq 1\}. \end{aligned}$$

Finite additivity and translation invariance give

$$m(V_t) = m(V).$$

The family indexed by $t \in [0, 1) \cap \mathbb{Q}$ is a partition of $[0, 1)$. Indeed, if $x \in [0, 1)$ and $y \in V$ represents the equivalence class of x , then either $t = x - y$ or $t = x - y + 1$ lies in $[0, 1) \cap \mathbb{Q}$ and $x \in V_t$. On the other hand, if $V_t \cap V_{t'} \neq \emptyset$, the uniqueness of representatives implies $t = t'$.

Countable additivity now yields

$$1 = m([0, 1)) = \sum_{t \in [0, 1) \cap \mathbb{Q}} m(V_t) = \sum_{t \in [0, 1) \cap \mathbb{Q}} m(V).$$

If $m(V) = 0$, the last sum is 0; if $m(V) > 0$, it is infinite. Either possibility is a contradiction. $\square$

One might try to rescue the construction by requiring only finite additivity together with invariance and normalization. In dimensions at least three, even this is impossible.

Theorem 2 (Banach–Tarski). *If $n \geq 3$, there is no finitely additive, rigid-motion-invariant map*

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

satisfying $m([0, 1]^n) = 1$.

The resolution is to give up the requirement that a measure be defined on *all* subsets of $\mathbb{R}^n$. In fact, measure theory can be developed on a general underlying set X once an appropriate collection of measurable subsets has been chosen.

1.2. Algebras and σ -algebras.

Definition 1 (Algebra). An algebra on a set X is a nonempty collection $\mathcal{A} \subseteq 2^X$ such that

- (a) if $E_1, \dots, E_N \in \mathcal{A}$, then $\bigcup_{j=1}^N E_j \in \mathcal{A}$;
- (b) if $E \in \mathcal{A}$, then $E^c = X \setminus E \in \mathcal{A}$.

Definition 2 (σ -algebra). A σ -algebra on a set X is a nonempty collection $\mathcal{A} \subseteq 2^X$ such that

- (a) if $E_n \in \mathcal{A}$ for every $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}$;
- (b) if $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.

Because the collections in these definitions are nonempty and closed under complements, they contain both X and $\emptyset$. By De Morgan's laws, an algebra is closed under finite intersections and a σ -algebra is closed under countable intersections. For example,

$$\bigcap_{n=1}^{\infty} E_n = \left(\bigcup_{n=1}^{\infty} E_n^c \right)^c.$$

When an algebra is already known to be closed under finite set operations, it is enough to check closure under countable *disjoint* unions in order to prove that it is a σ -algebra. Given $E_n \in \mathcal{A}$, set

$$\tilde{E}_1 = E_1, \quad \tilde{E}_n = E_n \setminus \bigcup_{j=1}^{n-1} E_j \quad (n \geq 2).$$

Then the sets $\tilde{E}_n$ are pairwise disjoint and

$$\bigcup_{n=1}^{\infty} \tilde{E}_n = \bigcup_{n=1}^{\infty} E_n.$$

Example 1. The collection $\{\emptyset, X\}$ is the smallest σ -algebra on X .

Example 2. The power set 2^X is the largest σ -algebra on X .

Example 3 (Countable-cocountable σ -algebra). The collection

$$\mathcal{A} = \{E \subseteq X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ -algebra on X .

A convenient way to specify a σ -algebra is to give a family of generators.

Definition 3 (Generated σ -algebra). *Let $\mathcal{E} \subseteq 2^X$. The σ -algebra generated by $\mathcal{E}$, denoted $\mathcal{M}(\mathcal{E})$, is the smallest σ -algebra containing $\mathcal{E}$.*

This object exists because 2^X is a σ -algebra containing $\mathcal{E}$, and an arbitrary intersection of σ -algebras is again a σ -algebra. Explicitly,

$$\mathcal{M}(\mathcal{E}) = \bigcap \{ \mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{E} \subseteq \mathcal{A} \}.$$

1.3. Borel σ -algebras.

Definition 4 (Borel σ -algebra). *Let X be a metric space, or more generally a topological space. The Borel σ -algebra $\mathcal{B}_X$ is the σ -algebra generated by the open subsets of X .*

For example, every open subset of $\mathbb{R}$ is a union of open intervals. The Borel σ -algebra contains every open and closed set, every G_δ set (a countable intersection of open sets), every F_σ set (a countable union of closed sets), and all sets obtained by continuing such countable operations. These classes form the Borel hierarchy.

Proposition 1. *The Borel σ -algebra $\mathcal{B}_{\mathbb{R}}$ is generated by any one of the following collections:*

- (a) $\{(a, b) : a < b\}$;
- (b) $\{[a, b] : a < b\}$;
- (c) $\{(-\infty, a] : a \in \mathbb{R}\}$.

Proof. Let the three collections above be $\mathcal{E}_1$, $\mathcal{E}_2$, and $\mathcal{E}_3$, respectively.

Every member of $\mathcal{E}_1$ is open, so $\mathcal{M}(\mathcal{E}_1) \subseteq \mathcal{B}_{\mathbb{R}}$. For the reverse inclusion, let $U \subseteq \mathbb{R}$ be open. For every $x \in U$, there is an $\varepsilon > 0$ such that

$$(x - \varepsilon, x + \varepsilon) \subseteq U.$$

By the density of $\mathbb{Q}$ in $\mathbb{R}$, we can choose $p, q \in \mathbb{Q}$ such that

$$x - \varepsilon < p < x < q < x + \varepsilon.$$

Thus every $x \in U$ lies in an interval (p, q) with rational endpoints that is contained in U . It follows that

$$U = \bigcup \{(p, q) : p, q \in \mathbb{Q}, p < q, (p, q) \subseteq U\}.$$

The indexing collection is a subset of $\mathbb{Q}^2$, so it is countable. Therefore every open subset of $\mathbb{R}$ is a countable union of open intervals. In particular, every open set belongs to $\mathcal{M}(\mathcal{E}_1)$, proving

$$\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}.$$

Every closed interval is Borel. In the other direction, for $a < b$,

$$(a, b) = \bigcup_{\substack{n \in \mathbb{N} \\ 2/n < b-a}} \left[a + \frac{1}{n}, b - \frac{1}{n} \right].$$

Thus $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_2)$, and consequently $\mathcal{M}(\mathcal{E}_2) = \mathcal{B}_{\mathbb{R}}$. Equivalently, the reverse containment can also be seen from

$$[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n} \right).$$

Finally, every left half-line in $\mathcal{E}_3$ is closed and hence Borel. Conversely,

$$(-\infty, b) = \bigcup_{n=1}^{\infty} \left(-\infty, b - \frac{1}{n} \right]$$

and therefore

$$(a, b) = (-\infty, b) \cap (-\infty, a]^c \in \mathcal{M}(\mathcal{E}_3).$$

It follows that $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_3)$, so $\mathcal{M}(\mathcal{E}_3) = \mathcal{B}_{\mathbb{R}}$ as well. $\square$

2. AUGUST 27, 2026

We continue our discussion of σ -algebras. Recall that a σ -algebra $\mathcal{M}$ on a set X contains $\emptyset$ and X , is closed under complements, and is closed under countable unions. If (X, d) is a metric space, then its Borel σ -algebra $\mathcal{B}_X$ is generated by the open subsets of X .

2.1. Product σ -algebras. Let I be an index set, let X_i be a nonempty set for each $i \in I$, and set

$$X = \prod_{i \in I} X_i.$$

For each $i \in I$, let $\mathcal{M}_i$ be a σ -algebra on X_i , and let

$$\pi_i: X \longrightarrow X_i, \quad \pi_i((x_j)_{j \in I}) = x_i,$$

be the coordinate projection.

Definition 5 (Product σ -algebra). *The product σ -algebra on X is*

$$\bigotimes_{i \in I} \mathcal{M}_i = \mathcal{M}(\{\pi_i^{-1}(E) : i \in I, E \in \mathcal{M}_i\}).$$

Thus it is the smallest σ -algebra that makes every coordinate projection measurable.

The generating sets $\pi_i^{-1}(E)$ are called *cylinder sets*. They restrict the i th coordinate to E and leave all other coordinates unrestricted.

Lemma 1. *If I is countable, then $\bigotimes_{i \in I} \mathcal{M}_i$ is generated by the measurable rectangles*

$$\left\{ \prod_{i \in I} E_i : E_i \in \mathcal{M}_i \right\}.$$

Proof. Since I is countable, every measurable rectangle satisfies

$$\prod_{i \in I} E_i = \bigcap_{i \in I} \pi_i^{-1}(E_i) \in \bigotimes_{i \in I} \mathcal{M}_i.$$

Conversely, each cylinder set is a measurable rectangle. Indeed,

$$\pi_j^{-1}(E_j) = \prod_{i \in I} F_i, \quad F_i = \begin{cases} E_j, & i = j, \\ X_i, & i \neq j. \end{cases}$$

The two collections therefore generate the same σ -algebra. $\square$

The generators of the coordinate σ -algebras can be used directly to generate the product.

Proposition 2. *Suppose*

$$\mathcal{M}_i = \mathcal{M}(\mathcal{G}_i), \quad \mathcal{G}_i \subseteq 2^{X_i},$$

for every $i \in I$.

(a) The product σ -algebra is generated by

$$\mathcal{F} = \{\pi_i^{-1}(G) : i \in I, G \in \mathcal{G}_i\}.$$

(b) If I is countable and $\mathcal{G}_i^* = \mathcal{G}_i \cup \{X_i\}$, then the product σ -algebra is generated by

$$\left\{ \prod_{i \in I} G_i : G_i \in \mathcal{G}_i^* \right\}.$$

Proof. For part (a), every member of $\mathcal{F}$ belongs to $\bigotimes_{i \in I} \mathcal{M}_i$, so

$$\mathcal{M}(\mathcal{F}) \subseteq \bigotimes_{i \in I} \mathcal{M}_i.$$

For the reverse inclusion, fix $i \in I$ and consider

$$\mathcal{A}_i = \{E \subseteq X_i : \pi_i^{-1}(E) \in \mathcal{M}(\mathcal{F})\}.$$

Preimages preserve complements and countable unions, so $\mathcal{A}_i$ is a σ -algebra. It contains $\mathcal{G}_i$ and hence contains $\mathcal{M}(\mathcal{G}_i) = \mathcal{M}_i$. Therefore every cylinder $\pi_i^{-1}(E)$ with $E \in \mathcal{M}_i$ belongs to $\mathcal{M}(\mathcal{F})$, which proves the reverse inclusion.

For part (b), each cylinder from part (a) is a rectangle whose other coordinates are X_j . Conversely, since I is countable, every rectangle in the displayed collection is the countable intersection of the sets $\pi_i^{-1}(G_i)$. When $G_i = X_i$ this set is all of X ; otherwise it is a cylinder from part (a). The conclusion follows. $\square$

2.2. Products of Borel σ -algebras. Let (X_i, d_i) , $i = 1, \dots, n$, be metric spaces. On

$$X = \prod_{i=1}^n X_i$$

define the maximum metric

$$d(x, y) = \max_{1 \leq i \leq n} d_i(x_i, y_i).$$

Proposition 3. *With the notation above,*

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X.$$

If every (X_i, d_i) is separable, then equality holds:

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} = \mathcal{B}_X.$$

Proof. Each coordinate projection $\pi_i: X \rightarrow X_i$ is continuous. If $U_i \subseteq X_i$ is open, then $\pi_i^{-1}(U_i)$ is open in X . Since $\mathcal{B}_{X_i}$ is generated by the open subsets of X_i , Proposition 2 gives

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X.$$

Now suppose that each X_i is separable, and choose a countable dense subset $D_i \subseteq X_i$. The collection

$$\mathcal{G}_i = \{B_{d_i}(p, q) : p \in D_i, q \in \mathbb{Q}, q > 0\}$$

is a countable basis for X_i and hence generates $\mathcal{B}_{X_i}$. The finite product $D = \prod_{i=1}^n D_i$ is countable and dense in X . Moreover,

$$B_d(x, q) = \prod_{i=1}^n B_{d_i}(x_i, q).$$

Thus every ball in X whose center lies in D and whose radius is a positive rational belongs to $\bigotimes_{i=1}^n \mathcal{B}_{X_i}$. These balls form a countable basis for X , so every open subset of X is a countable union of such balls. Hence

$$\mathcal{B}_X \subseteq \bigotimes_{i=1}^n \mathcal{B}_{X_i},$$

which completes the proof. $\square$

Corollary 1. *For every $n \in \mathbb{N}$,*

$$\mathcal{B}_{\mathbb{R}^n} = \bigotimes_{i=1}^n \mathcal{B}_{\mathbb{R}}.$$

For example, $\mathcal{B}_{\mathbb{R}^n}$ is generated by each of the following collections:

- open rectangles $\prod_{i=1}^n (a_i, b_i)$;

- closed rectangles $\prod_{i=1}^n [a_i, b_i]$; and
- lower orthants $\prod_{i=1}^n (-\infty, a_i]$.

2.3. Measures.

Definition 6 (Measure). *Let $\mathcal{M}$ be a σ -algebra on a set X . A function*

$$\mu: \mathcal{M} \longrightarrow [0, \infty]$$

is a measure if

- (i) $\mu(\emptyset) = 0$;
- (ii) *whenever $E_1, E_2, \dots \in \mathcal{M}$ are pairwise disjoint,*

$$\mu \left(\bigcup_{i=1}^{\infty} E_i \right) = \sum_{i=1}^{\infty} \mu(E_i).$$

The second property is called countable additivity. If it is required only for finite pairwise disjoint families, then μ is called finitely additive.

The sets in $\mathcal{M}$ are called *measurable sets*, and the triple $(X, \mathcal{M}, \mu)$ is called a *measure space*. A measure μ is

- *finite* if $\mu(X) < \infty$;
- σ -*finite* if $X = \bigcup_{i=1}^{\infty} E_i$ for some $E_i \in \mathcal{M}$ with $\mu(E_i) < \infty$ for every i .

A set $N \in \mathcal{M}$ with $\mu(N) = 0$ is called a *null set*. A statement is said to hold *almost everywhere*, abbreviated a.e., if it holds outside some null set.

2.4. Examples of measures.

Example 4 (Weighted counting measure). *Let X be nonempty, let $\mathcal{M} = 2^X$, and let $f: X \rightarrow [0, \infty]$. Define*

$$\mu(E) = \sum_{x \in E} f(x) := \sup \left\{ \sum_{x \in F} f(x) : F \subseteq E, F \text{ finite} \right\}.$$

Then μ is a measure on 2^X .

If $f(x) = 1$ for every $x \in X$, then μ is the counting measure,

$$\mu(E) = \#E.$$

If $x_0 \in X$ and

$$f(x) = \begin{cases} 1, & x = x_0, \\ 0, & x \neq x_0, \end{cases}$$

then

$$\mu(E) = \begin{cases} 1, & x_0 \in E, \\ 0, & x_0 \notin E. \end{cases}$$

This is the point mass, or Dirac measure, at x_0 .

Example 5 (Countable-cocountable measure). Let X be uncountable, and let $\mathcal{M}$ be the countable-cocountable σ -algebra. Then

$$\mu(E) = \begin{cases} 0, & E \text{ is countable,} \\ 1, & E^c \text{ is countable} \end{cases}$$

defines a measure on $(X, \mathcal{M})$.

Example 6 (Finite additivity is not enough). Let $X = \mathbb{N}$ and $\mathcal{M} = 2^{\mathbb{N}}$. Define

$$\nu(E) = \begin{cases} 0, & E \text{ is finite,} \\ \infty, & E \text{ is infinite.} \end{cases}$$

Then ν is finitely additive. It is not a measure, because

$$\nu\left(\bigcup_{n=1}^{\infty} \{n\}\right) = \infty \quad \text{but} \quad \sum_{n=1}^{\infty} \nu(\{n\}) = 0.$$

2.5. Basic properties of measures.

Theorem 3. Let $(X, \mathcal{M}, \mu)$ be a measure space.

(a) Monotonicity: if $E, F \in \mathcal{M}$ and $E \subseteq F$, then

$$\mu(E) \leq \mu(F).$$

(b) Countable subadditivity: if $E_1, E_2, \dots \in \mathcal{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mu(E_i).$$

(c) Continuity from below: if

$$E_1 \subseteq E_2 \subseteq \dots, \quad E_i \in \mathcal{M},$$

then

$$\mu \left(\bigcup_{i=1}^{\infty} E_i \right) = \lim_{i \rightarrow \infty} \mu(E_i).$$

(d) Continuity from above: if

$$E_1 \supseteq E_2 \supseteq \cdots, \quad E_i \in \mathcal{M}, \quad \mu(E_1) < \infty,$$

then

$$\mu \left(\bigcap_{i=1}^{\infty} E_i \right) = \lim_{i \rightarrow \infty} \mu(E_i).$$

Proof. For part (a), write F as the disjoint union

$$F = E \cup (F \setminus E).$$

Countable additivity gives

$$\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E).$$

For part (b), define

$$F_1 = E_1, \quad F_k = E_k \setminus \bigcup_{j=1}^{k-1} E_j \quad (k \geq 2).$$

The sets F_k are pairwise disjoint, $F_k \subseteq E_k$, and

$$\bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} E_k.$$

Therefore countable additivity and part (a) imply

$$\mu \left(\bigcup_{k=1}^{\infty} E_k \right) = \sum_{k=1}^{\infty} \mu(F_k) \leq \sum_{k=1}^{\infty} \mu(E_k).$$

For part (c), set

$$F_1 = E_1, \quad F_k = E_k \setminus E_{k-1} \quad (k \geq 2).$$

The sets F_k are pairwise disjoint, and for each N ,

$$E_N = \bigcup_{k=1}^N F_k, \quad \bigcup_{i=1}^{\infty} E_i = \bigcup_{k=1}^{\infty} F_k.$$

It follows that

$$\begin{aligned}\mu\left(\bigcup_{i=1}^{\infty} E_i\right) &= \sum_{k=1}^{\infty} \mu(F_k) \\ &= \lim_{N \rightarrow \infty} \sum_{k=1}^N \mu(F_k) \\ &= \lim_{N \rightarrow \infty} \mu(E_N).\end{aligned}$$

For part (d), define

$$F_i = E_1 \setminus E_i.$$

Since E_i decreases, F_i increases, and

$$\bigcup_{i=1}^{\infty} F_i = E_1 \setminus \bigcap_{i=1}^{\infty} E_i.$$

By part (c),

$$\mu\left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_1 \setminus E_i).$$

Because $E_i \subseteq E_1$ and $\mu(E_1) < \infty$, finite additivity gives

$$\mu(E_1 \setminus E_i) = \mu(E_1) - \mu(E_i)$$

and

$$\mu\left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i\right) = \mu(E_1) - \mu\left(\bigcap_{i=1}^{\infty} E_i\right).$$

Substituting these identities into the preceding limit and cancelling the finite quantity $\mu(E_1)$ yields

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i).$$

□

Remark 1 (Necessity of the finite-measure hypothesis). The assumption $\mu(E_1) < \infty$ in Theorem 3(d) cannot be omitted. Let

$$X = \mathbb{N}, \quad \mathcal{M} = 2^{\mathbb{N}},$$

let μ be counting measure, and define

$$E_i = \{i, i+1, i+2, \dots\}.$$

Then

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \dots, \quad \bigcap_{i=1}^{\infty} E_i = \emptyset.$$

Indeed, for any $n \in \mathbb{N}$, we have $n \notin E_i$ whenever $i > n$. Consequently,

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \mu(\emptyset) = 0.$$

On the other hand, every E_i is infinite, so

$$\mu(E_i) = \infty \quad \text{for every } i, \quad \lim_{i \rightarrow \infty} \mu(E_i) = \infty.$$

Thus

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = 0 \neq \infty = \lim_{i \rightarrow \infty} \mu(E_i).$$

This does not contradict part (d), because $\mu(E_1) = \mu(\mathbb{N}) = \infty$.

3. SEPTEMBER 1, 2026

Recall that a measure on a σ -algebra $\mathcal{M}$ over a set X is a map

$$\mu: \mathcal{M} \longrightarrow [0, \infty]$$

such that $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

whenever $A_1, A_2, \dots \in \mathcal{M}$ are pairwise disjoint. For Lebesgue measure m on $\mathbb{R}$, the normalization is $m([0, 1]) = 1$.

3.1. Complete measure spaces.

Definition 7 (Complete measure). *A measure μ on a σ -algebra $\mathcal{M}$ over X is complete if every subset of a null set is measurable. That is, if $E \subseteq N$ for some $N \in \mathcal{M}$ with $\mu(N) = 0$, then $E \in \mathcal{M}$.*

By monotonicity, every such set E also satisfies

$$0 \leq \mu(E) \leq \mu(N) = 0,$$

so $\mu(E) = 0$.

Every measure space has a canonical completion obtained by adjoining all subsets of its null sets.

Proposition 4 (Completion of a measure). *Let $(X, \mathcal{M}, \mu)$ be a measure space, and let*

$$\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}.$$

Define

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M}, F \subseteq N \text{ for some } N \in \mathcal{N}\}.$$

Then the following statements hold.

- (i) *The collection $\overline{\mathcal{M}}$ is a σ -algebra on X containing $\mathcal{M}$.*
- (ii) *There is a unique measure $\bar{\mu}$ on $\overline{\mathcal{M}}$ extending μ . It is given by*

$$\bar{\mu}(E \cup F) = \mu(E),$$

where $E \in \mathcal{M}$ and F is contained in a member of $\mathcal{N}$. Moreover, $\bar{\mu}$ is complete.

Proof. The collections $\mathcal{M}$ and $\mathcal{N}$ are closed under countable unions. For $\mathcal{N}$, this follows from countable subadditivity: if $N_i \in \mathcal{N}$, then

$$0 \leq \mu\left(\bigcup_{i=1}^{\infty} N_i\right) \leq \sum_{i=1}^{\infty} \mu(N_i) = 0.$$

Hence $\overline{\mathcal{M}}$ is closed under countable unions. Indeed, if $E_i \in \mathcal{M}$ and $F_i \subseteq N_i \in \mathcal{N}$, then

$$\bigcup_{i=1}^{\infty} (E_i \cup F_i) = \left(\bigcup_{i=1}^{\infty} E_i\right) \cup \left(\bigcup_{i=1}^{\infty} F_i\right),$$

and $\bigcup_i F_i \subseteq \bigcup_i N_i \in \mathcal{N}$.

To check closure under complements, let $E \in \mathcal{M}$ and $F \subseteq N \in \mathcal{N}$. Replacing N by $N \setminus E$ and F by $F \setminus E$ does not change $E \cup F$, so we

may assume that $E \cap N = \emptyset$. In that case,

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F).$$

The first set on the right belongs to $\mathcal{M}$, and the second is contained in the null set N . Thus $(E \cup F)^c$ belongs to $\overline{\mathcal{M}}$. Since $\emptyset \in \overline{\mathcal{M}}$, this proves that $\overline{\mathcal{M}}$ is a σ -algebra. Taking $F = \emptyset$ shows that $\mathcal{M} \subseteq \overline{\mathcal{M}}$.

We next show that the formula for $\bar{\mu}$ is well defined. Suppose

$$E_1 \cup F_1 = E_2 \cup F_2, \quad E_i \in \mathcal{M}, \quad F_i \subseteq N_i \in \mathcal{N}.$$

Then $E_1 \subseteq E_2 \cup N_2$, and therefore

$$\mu(E_1) \leq \mu(E_2 \cup N_2) \leq \mu(E_2) + \mu(N_2) = \mu(E_2).$$

By symmetry, $\mu(E_2) \leq \mu(E_1)$, so $\mu(E_1) = \mu(E_2)$.

Clearly $\bar{\mu}(\emptyset) = 0$. Suppose that $A_i = E_i \cup F_i$ are pairwise disjoint members of $\overline{\mathcal{M}}$, where $E_i \in \mathcal{M}$ and $F_i \subseteq N_i \in \mathcal{N}$. The sets E_i are pairwise disjoint, so

$$\begin{aligned} \bar{\mu}\left(\bigcup_{i=1}^{\infty} A_i\right) &= \mu\left(\bigcup_{i=1}^{\infty} E_i\right) \\ &= \sum_{i=1}^{\infty} \mu(E_i) \\ &= \sum_{i=1}^{\infty} \bar{\mu}(A_i). \end{aligned}$$

Thus $\bar{\mu}$ is a measure, and it extends μ .

If $A = E \cup F \in \overline{\mathcal{M}}$ and $\bar{\mu}(A) = 0$, then $\mu(E) = 0$. Since $A \subseteq E \cup N$ and $E \cup N$ is a measurable null set, every subset of A belongs to $\overline{\mathcal{M}}$. Hence $\bar{\mu}$ is complete.

Finally, let ν be any measure on $\overline{\mathcal{M}}$ that extends μ . If $A = E \cup F$, with $E \in \mathcal{M}$ and $F \subseteq N \in \mathcal{N}$, then

$$E \subseteq A \subseteq E \cup N.$$

Monotonicity and the fact that ν extends μ give

$$\mu(E) = \nu(E) \leq \nu(A) \leq \nu(E \cup N) = \mu(E \cup N) = \mu(E).$$

Thus $\nu(A) = \mu(E) = \bar{\mu}(A)$, which proves uniqueness. $\square$

The measure $\bar{\mu}$ is called the *completion* of μ .

3.2. Outer measures. The goal is to construct measures from simple geometric pieces. For example, to estimate the area of a set $E \subseteq \mathbb{R}^2$, one can cover E by squares of side length s and consider

$$\inf_{s>0} (\text{number of squares in the cover}) s^2.$$

More generally, one covers a set by countably many simple pieces and takes the infimum of the sums of their assigned sizes.

A related guiding principle is that the “reasonable” subsets $E \subseteq [0, 1]^2$ should be those for which outer and inner area agree. If the unit square has area 1, its inner estimate for E is

$$1 - m^*([0, 1]^2 \setminus E).$$

Definition 8 (Outer measure). *An outer measure on a set X is a map*

$$\mu^*: 2^X \longrightarrow [0, \infty]$$

satisfying the following properties.

- (i) $\mu^*(\emptyset) = 0$.
- (ii) If $A \subseteq B \subseteq X$, then $\mu^*(A) \leq \mu^*(B)$.
- (iii) For every sequence $A_1, A_2, \dots \subseteq X$,

$$\mu^*\left(\bigcup_{j=1}^{\infty} A_j\right) \leq \sum_{j=1}^{\infty} \mu^*(A_j).$$

Unlike a measure, an outer measure is defined on every subset of X ; its countable subadditivity therefore applies without a measurability assumption.

Proposition 5 (Outer measure from coverings). *Let $\mathcal{E} \subseteq 2^X$ contain $\emptyset$ and X , and let*

$$\rho: \mathcal{E} \longrightarrow [0, \infty]$$

satisfy $\rho(\emptyset) = 0$. For $A \subseteq X$, define

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E}, \quad A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}.$$

Then μ^* is an outer measure on X .

Remark 2. Even when $E \in \mathcal{E}$, it need not be true that $\mu^*(E) = \rho(E)$. A more efficient countable cover may have smaller total ρ -size than the single-set cover by E .

Example 7. Let $X = \mathbb{R}$ and take

$$\mathcal{E} = \{\emptyset, \mathbb{R}\} \cup \{[a, b] : a, b \in \mathbb{R}, a \leq b\}.$$

Define

$$\rho(\emptyset) = 0, \quad \rho(\mathbb{R}) = \infty, \quad \rho([a, b]) = (b - a)^2.$$

Although $\rho([0, 1]) = 1$, the induced outer measure satisfies $\mu^*([0, 1]) = 0$. Indeed, for every $n \in \mathbb{N}$,

$$[0, 1] = \bigcup_{j=0}^{n-1} \left[\frac{j}{n}, \frac{j+1}{n} \right],$$

and the total ρ -size of this cover is

$$\sum_{j=0}^{n-1} \left(\frac{1}{n} \right)^2 = \frac{1}{n}.$$

Hence $0 \leq \mu^*([0, 1]) \leq 1/n$ for every n , which gives the claim.

Proof of the proposition. The defining collection of sums is nonempty because any set $A \subseteq X$ is covered by the sequence $X, \emptyset, \emptyset, \dots$. Thus $\mu^*(A)$ is well defined as an element of $[0, \infty]$.

The sequence $\emptyset, \emptyset, \dots$ covers $\emptyset$, so

$$0 \leq \mu^*(\emptyset) \leq \sum_{j=1}^{\infty} \rho(\emptyset) = 0.$$

If $A \subseteq B$, then every countable cover of B by members of $\mathcal{E}$ is also a cover of A . Taking infima over the respective collections of covers gives $\mu^*(A) \leq \mu^*(B)$.

It remains to prove countable subadditivity. Let $A_j \subseteq X$ for $j \in \mathbb{N}$. If $\sum_j \mu^*(A_j) = \infty$, there is nothing to prove. Otherwise, fix $\varepsilon > 0$. For every j , choose $E_{j,k} \in \mathcal{E}$, $k \in \mathbb{N}$, such that

$$A_j \subseteq \bigcup_{k=1}^{\infty} E_{j,k} \quad \text{and} \quad \sum_{k=1}^{\infty} \rho(E_{j,k}) \leq \mu^*(A_j) + \varepsilon 2^{-j}.$$

The doubly indexed collection $\{E_{j,k} : j, k \in \mathbb{N}\}$ is countable and covers $\bigcup_j A_j$. Consequently,

$$\begin{aligned} \mu^*\left(\bigcup_{j=1}^{\infty} A_j\right) &\leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \rho(E_{j,k}) \\ &\leq \sum_{j=1}^{\infty} (\mu^*(A_j) + \varepsilon 2^{-j}) \\ &= \sum_{j=1}^{\infty} \mu^*(A_j) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ proves the result. $\square$

Definition 9 (Measurability with respect to an outer measure). *Let μ^* be an outer measure on X . A set $A \subseteq X$ is μ^* -measurable if, for every $E \subseteq X$,*

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Countable subadditivity always gives

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c),$$

so the content of the measurability criterion is the reverse inequality.

When $\mu^*(E) < \infty$, the quantity

$$\mu^*(E) - \mu^*(E \cap A^c)$$

can be viewed as the inner measure of $E \cap A$ relative to E .

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Recall that if μ^* is an outer measure on X , a set $A \subseteq X$ is μ^* -measurable if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for every $E \subseteq X$. By countable subadditivity, it is enough to prove the inequality

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

4.1. The Carathéodory construction.

Theorem 4 (Carathéodory). *Let μ^* be an outer measure on X , and set*

$$\mathcal{M}^* = \{A \subseteq X : A \text{ is } \mu^*\text{-measurable}\}.$$

Then $\mathcal{M}^$ is a σ -algebra, and*

$$\mu = \mu^*|_{\mathcal{M}^*}$$

is a complete measure on $\mathcal{M}^$.*

Proof. We first show that $\mathcal{M}^*$ is an algebra. The empty set is measurable because

$$\mu^*(E) = \mu^*(E \cap \emptyset) + \mu^*(E \cap X)$$

for every $E \subseteq X$. The defining condition is symmetric in A and A^c , so

$$A \in \mathcal{M}^* \implies A^c \in \mathcal{M}^*.$$

Now let $A, B \in \mathcal{M}^*$ and fix $E \subseteq X$. Applying the measurability criterion first to A and then to B gives

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) \\ &\quad + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c). \end{aligned}$$

The first three sets on the last two lines have union $E \cap (A \cup B)$. Therefore countable subadditivity implies

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c).$$

Thus $A \cup B \in \mathcal{M}^*$, and $\mathcal{M}^*$ is an algebra.

This argument also establishes finite additivity on disjoint measurable sets. Indeed, if $A, B \in \mathcal{M}^*$ are disjoint, apply the measurability of A with $E = A \cup B$ to obtain

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B).$$

To prove closure under countable unions, first let $A_1, A_2, \dots \in \mathcal{M}^*$ be pairwise disjoint, and set

$$B = \bigcup_{i=1}^{\infty} A_i, \quad B_n = \bigcup_{i=1}^n A_i.$$

Since $\mathcal{M}^*$ is an algebra, $B_n \in \mathcal{M}^*$. Repeatedly applying the measurability criterion and using the disjointness of the A_i gives, for every $E \subseteq X$,

$$\mu^*(E) = \sum_{i=1}^n \mu^*(E \cap A_i) + \mu^*(E \cap B_n^c).$$

Because $B^c \subseteq B_n^c$, monotonicity yields

$$\mu^*(E) \geq \sum_{i=1}^n \mu^*(E \cap A_i) + \mu^*(E \cap B^c).$$

Letting $n \rightarrow \infty$ and then using countable subadditivity, we find

$$\begin{aligned} \mu^*(E) &\geq \sum_{i=1}^{\infty} \mu^*(E \cap A_i) + \mu^*(E \cap B^c) \\ &\geq \mu^*\left(\bigcup_{i=1}^{\infty} (E \cap A_i)\right) + \mu^*(E \cap B^c) \\ &= \mu^*(E \cap B) + \mu^*(E \cap B^c). \end{aligned}$$

Hence $B \in \mathcal{M}^*$. An arbitrary countable union can be disjointified by replacing A_i with

$$A_i \setminus \bigcup_{j=1}^{i-1} A_j.$$

It follows that $\mathcal{M}^*$ is a σ -algebra.

The same computation proves countable additivity. If the measurable sets A_i are pairwise disjoint and $B = \bigcup_i A_i$, then countable subadditivity gives

$$\mu^*(B) \leq \sum_{i=1}^{\infty} \mu^*(A_i).$$

On the other hand, for every n ,

$$\mu^*(B) = \sum_{i=1}^n \mu^*(A_i) + \mu^*(B \setminus B_n) \geq \sum_{i=1}^n \mu^*(A_i).$$

Letting $n \rightarrow \infty$ proves

$$\mu^*(B) = \sum_{i=1}^{\infty} \mu^*(A_i).$$

Thus $\mu = \mu^*|_{\mathcal{M}^*}$ is a measure.

Finally, suppose that $A \subseteq X$ and $\mu^*(A) = 0$. For every $E \subseteq X$, subadditivity and monotonicity give

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq 0 + \mu^*(E) = \mu^*(E).$$

Hence $A \in \mathcal{M}^*$. In particular, if $N \in \mathcal{M}^*$ satisfies $\mu(N) = 0$, then every subset $A \subseteq N$ has outer measure zero and therefore belongs to $\mathcal{M}^*$. This proves that μ is complete. $\square$

4.2. Premeasures. The covering construction from the preceding lecture begins with a set function ρ defined on a family of elementary sets. The example $\rho([a, b]) = (b - a)^2$ showed that the resulting outer measure may fail to agree with ρ , even on elementary sets. To prevent this collapse, we require the initial set function to behave like a measure.

Definition 10 (Premeasure). *Let $\mathcal{A}$ be an algebra on X . A map*

$$\mu_0: \mathcal{A} \longrightarrow [0, \infty]$$

is a premeasure if

$$(i) \mu_0(\emptyset) = 0;$$

(ii) if $A_1, A_2, \dots \in \mathcal{A}$ are pairwise disjoint and $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$, then

$$\mu_0 \left(\bigcup_{i=1}^{\infty} A_i \right) = \sum_{i=1}^{\infty} \mu_0(A_i).$$

Thus a premeasure has the same additivity property as a measure, but its domain is only required to be an algebra, rather than a σ -algebra.

Example 8 (Half-open intervals). Let $X = \mathbb{R}$. Consider intervals of the forms

$$(a, b], \quad (-\infty, b], \quad (a, \infty),$$

together with $\emptyset$, where $a, b \in \mathbb{R}$ and $a < b$. The collection $\mathcal{A}$ of finite unions of such intervals is an algebra. Moreover,

$$\mathcal{M}(\mathcal{A}) = \mathcal{B}_{\mathbb{R}}.$$

This algebra will later be used to construct Lebesgue measure.

The next proposition identifies the hypotheses under which the outer measure construction preserves the initial set function.

Proposition 6 (Extension of a premeasure). Let μ_0 be a premeasure on an algebra $\mathcal{A}$ over X . For every $E \subseteq X$, define

$$\mu^*(E) = \inf \left\{ \sum_{i=1}^{\infty} \mu_0(A_i) : A_i \in \mathcal{A}, \quad E \subseteq \bigcup_{i=1}^{\infty} A_i \right\}.$$

Then the following statements hold.

(i) $\mu^*|_{\mathcal{A}} = \mu_0$.

(ii) Every member of $\mathcal{A}$ is μ^* -measurable.

Consequently, $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{M}^*$, and the measure $\mu^*|_{\mathcal{M}(\mathcal{A})}$ extends μ_0 .

Proof. Let $A \in \mathcal{A}$. The sequence $A, \emptyset, \emptyset, \dots$ is a cover of A , so

$$\mu^*(A) \leq \mu_0(A).$$

Conversely, suppose that $A_i \in \mathcal{A}$ and $A \subseteq \bigcup_{i=1}^{\infty} A_i$. Define

$$B_1 = A \cap A_1, \quad B_k = A \cap \left(A_k \setminus \bigcup_{i=1}^{k-1} A_i \right) \quad (k \geq 2).$$

Then the sets B_k are pairwise disjoint members of $\mathcal{A}$, and

$$A = \bigcup_{k=1}^{\infty} B_k.$$

Since $B_k \subseteq A_k$, monotonicity of the premeasure gives

$$\mu_0(A) = \sum_{k=1}^{\infty} \mu_0(B_k) \leq \sum_{k=1}^{\infty} \mu_0(A_k).$$

Taking the infimum over all such covers yields $\mu_0(A) \leq \mu^*(A)$. Therefore $\mu^*(A) = \mu_0(A)$, proving part (i).

For part (ii), fix $A \in \mathcal{A}$ and $E \subseteq X$. If $\mu^*(E) = \infty$, the desired measurability identity follows at once from countable subadditivity. Suppose instead that $\mu^*(E) < \infty$, and let $\varepsilon > 0$. Choose $A_1, A_2, \dots \in \mathcal{A}$ such that

$$E \subseteq \bigcup_{i=1}^{\infty} A_i \quad \text{and} \quad \mu^*(E) + \varepsilon \geq \sum_{i=1}^{\infty} \mu_0(A_i).$$

Since μ_0 is finitely additive,

$$\begin{aligned} \sum_{i=1}^{\infty} \mu_0(A_i) &= \sum_{i=1}^{\infty} (\mu_0(A_i \cap A) + \mu_0(A_i \cap A^c)) \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c). \end{aligned}$$

Hence

$$\mu^*(E) + \varepsilon \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Letting $\varepsilon \downarrow 0$ proves that A is μ^* -measurable.

By Theorem 4, $\mathcal{M}^*$ is a σ -algebra. Since it contains $\mathcal{A}$, it also contains $\mathcal{M}(\mathcal{A})$. Part (i) then shows that the restriction of μ^* to $\mathcal{M}(\mathcal{A})$ extends μ_0 . $\square$